

Cybersecurity Attack Types Study Guide

Network & Communication Attacks

Password Attacks

A password attack happens when a hacker tries to steal or guess a password to get into a locked device, network, or account.

- **Brute Force:** The hacker uses a fast computer program to automatically guess millions of password combinations (like a, b, c, 1, 2, 3) until it finds the correct one. It is like trying every single key on a massive keyring until one opens the door.
- **Rainbow Table:** Hackers use a pre-made cheat sheet called a "rainbow table." This list contains millions of pre-calculated, encrypted passwords. Instead of guessing from scratch, the hacker compares the stolen encrypted data to this list to quickly reveal the real password.

***Security Domain:** Communication and Network Security domain.*

Cryptographic Attacks

Cryptography is the science of locking text behind secret codes (encryption) so only the intended receiver can read it. A cryptographic attack is an attempt to break these codes and spy on messages.

- **Birthday Attack:** This attack is based on a math puzzle called the "birthday paradox" (which proves that in a small room of people, it is surprisingly easy to find two people with the exact same birthday). Hackers use this math to find two different files that create the exact same digital code, allowing them to swap a safe file with a virus.
- **Collision:** A collision happens when two completely different pieces of data accidentally create the exact same digital signature. Hackers hunt for these rare glitches to fool a system into accepting a fake document as a real one.
- **Downgrade Attack:** The hacker forces a computer network to stop using its modern, high-tech security and drop down to an older, weaker version of security. Once the system is using the weak version, it becomes much easier to hack.

***Security Domain:** Communication and Network Security domain.*

Human & Manipulation Attacks (Social Engineering)

Social engineering is the art of manipulating and tricking human beings. Instead of hacking a computer with code, the criminal hacks a person's trust, curiosity, or fear to make them give away secrets.

The Phishing Family

- **Phishing:** Fake emails sent out to thousands of random people at once. The email pretends to be from a trusted bank or company, asking you to click a link and type in your password.
- **Smishing:** The exact same trick as phishing, but done through SMS text messages on your mobile phone instead of email.

- **Vishing:** Voice phishing. This is when a scammer calls you on the phone pretending to be someone official—like a bank agent or the police—to scare you into giving them money or account details.
- **Spear Phishing:** A highly personalized attack. Instead of emailing random people, the hacker targets one specific person. They research you first (using your social media) so the fake email mentions your real boss, your real project, or your real hobbies to look completely believable.
- **Whaling:** Spear phishing aimed only at "big fish," like CEOs, company executives, or government leaders. The emails look like legal threats or major business deals to trick high-level bosses into transferring millions of dollars.
- **Social Media Phishing:** Scams that happen inside apps like Instagram, Facebook, or LinkedIn. Hackers send fake direct messages (DMs) or post fake links pretending to offer customer support or viral videos to steal your login details.

Other Social Engineering Tricks

- **Business Email Compromise (BEC):** A hacker secretly breaks into a real manager's corporate email account. They then send an email from that real account to the finance department, saying, "Please pay this vendor invoice immediately." Because the email is real, employees get tricked into sending money straight to the hacker.
- **Watering Hole Attack:** A hacker finds a website that a specific target group visits very often (like a local neighborhood forum or a niche industry blog). The hacker infects that specific website with malware, waiting for the targets to visit and automatically catch the virus.
- **USB Baiting:** A hacker leaves a virus-infected USB flash drive lying on the ground in a public place, like a corporate parking lot or lobby. They might write a label on it like "Salary Details." A curious employee picks it up, plugs it into their work computer to see what is on it, and accidentally infects the whole company network.
- **Physical Social Engineering:** Tricking people face-to-face in the physical world. For example, a hacker might dress up as a plumber or a delivery driver, hold a heavy box, and ask an employee to hold the secure office door open for them (a trick called "tailgating").

Security Domain: Security and Risk Management domain.

Physical & Hardware Attacks

A physical attack happens when a hacker actually touches or tampers with physical objects and hardware in the real world to compromise security.

- **Malicious USB Cable:** A charging cable that looks completely normal on the outside but contains a hidden microchip on the inside. When you plug your phone or laptop into it to charge, the hidden chip secretly types commands into your computer to steal files.
- **Malicious Flash Drive:** A thumb drive pre-loaded with hidden malicious software. The moment it enters a computer's slot, it can automatically act like a keyboard, typing invisible commands to take over the operating system.
- **Card Cloning and Skimming:** Scammers attach small, hidden plastic readers (skimmers) over the card slots of real ATMs or gas pumps. When you slide your credit card in, the device reads your magnetic strip. The hacker copies that data onto a blank plastic card (cloning) to go shopping with your money.

Security Domain: Asset Security domain.

Advanced & Process Attacks

Adversarial Artificial Intelligence

This happens when hackers turn AI and machine learning against itself. They can feed corrupt or misleading data into an AI system during its training phase to create a "blind spot." Once the AI goes live, the hacker can pass right by it without being detected. Hackers also use AI tools to generate highly realistic fake videos (deepfakes) or write perfect phishing emails automatically.

Security Domain: *Communication and Network Security, and Identity and Access Management domains.*

Supply-Chain Attack

Products are rarely built by just one company; they rely on a chain of third-party suppliers, software developers, and shipping companies. In a supply-chain attack, the hacker does not attack the main target directly. Instead, they hack a smaller supplier who makes a tiny piece of the target's software or hardware. By hiding a virus inside that small component, the virus gets delivered straight inside the secure target organization automatically.

Security Domain: *Security and Risk Management, Security Architecture and Engineering, and Security Operations domains.*